AI Interaction Behavior Mapping

From User Action to Planetary Consequence

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Rapid Iterative Prompting

User Behavior

Sending multiple consecutive prompts to refine AI outputs within minutes

Commonality

Very common in creative work, debugging, and exploratory tasks

Frictionless Feel

Instant responses create illusion of zero cost interaction

BEHAVIOR → CONSEQUENCE CHAIN

User Layer

- **Query Burst** — 3-10 prompts/min
- **Visible** — Instant feedback
- **Temporal Scale** — Session-based

Infrastructure

- **GPU Cluster Spike** — Each query = thousands of GPUs
- **Data Center Load** — Peak power draw
- **Network Traffic** — Multi-region routing
- **Invisible to User** — Abstracted away

Planetary Impact

- **CO_2 Emissions** — ~0.5kg per iteration
- **Water Usage** — 500ml cooling/query
- **Heat Dissipation** — Urban heat islands
- **Invisible Impact** — Accumulative

PATTERN DYNAMICS MODEL

Intensity Over Time

Resource Load — Session-based spike · Infrastructure cascade · Permanent accumulation · User-invisible process

User query spike

Idle/decay period

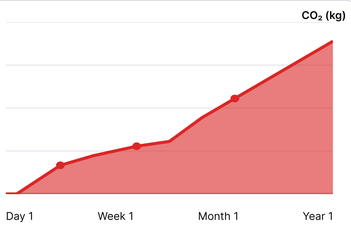
What Triggers Escalation

Iterative refinement loops - each unsatisfactory output prompts another query, creating cascade effect

Why This Remains Invisible

Infrastructure abstracted behind instant UI response; no feedback on resource consumption per query

Cumulative Impact



Dynamic: Accumulation

No reset mechanism - environmental cost compounds over time

How Intensity Evolves

Spike during active sessions, rapid decay when idle, but planetary impact accumulates permanently

Design Material Potential

Visualizing cumulative impact could inform conscious usage patterns and prompt batching strategies